Luyao.Derek. Xia

AI Specilist

38A Schnapper Rock Road, Schnapper Rock, Auckland, New Zealand, 0632
+64 2040007197
luyao.xia01@gmail.com

Feb 1, 1993
New Zealand

AI Researcher with a PhD about applying artificial intelligence to real-world challenges. With a strong background in computer vision, generative AI, machine learning, and the application of large language models and AI Agents, I specialize in developing machine learning models and embedding AI-powered solutions into workflows. I am excited about leveraging advanced AI techniques to transform business operations and drive future innovations.

Employment history

Postdoctoral Fellow, Auckland Bioengineering Institute, University of Auckland, New Zealand, Jul 2025 - Present

Auckland, New Zealand

- **Responsibilities:**
  - Lead innovative research in intelligent perception and human–AI collaboration, fostering PhD student development.
  - Developed 3D semantic segmentation algorithms to model and track soft tissue deformation during surgery using computer vision.
  - Collaborated with industry partners on AI Agent for Smart Glasses, integrating AI and physiological signals for real-time health management on wearable devices.
  - Authored high-impact publications and secured research grants, strengthening institutional research visibility.

  Major Achievements:
  - Implemented a real-time 3D surgical segmentation pipeline enabling precise soft-tissue tracking and intraoperative monitoring.
  - Designed and deployed a multi-agent AI architecture for smart glasses, combining spatial vision and physiological sensing for context-aware, adaptive health feedback.
  - Advanced translational research bridging biomedical engineering, AI, and human-computer interaction.
  - Mentored postgraduate students and coordinated academic-industry collaborations to drive innovation.

AI Engineer (Internship), China UnionPay Electronic Payment Research Institute, Shanghai, Jun 2019 - Dec 2019

Shanghai, China

- **Responsibilities:**
  - Participated in the AI-driven non-inductive car payment project, integrating computer vision and backend payment systems.
  - Developed license plate recognition models to enable automated vehicle identification and seamless transaction processing.
  - Built and deployed a non-inductive payment server using Spring Boot, supporting real-time transaction validation and scalability.

  Major Achievements:
  - Applied computer vision-based recognition and AI tools to streamline vehicle payment workflows, significantly improving operational efficiency.
  - Enhanced transaction automation and reliability through the integration of AI inference pipelines into production systems.
  - Contributed to the prototype design of a smart transportation payment ecosystem, later used for corporate R&D demonstrations.

Employment history

Data Engineer (Internship), China UnionPay, Shanghai, Jul 2018 - Oct 2018
Shanghai, China

- **Responsibilities:**
  - Designed and implemented database partition strategies to enhance data management and performance.
  - Developed and tested database optimization algorithms to improve query efficiency and system scalability.

  Major Achievements:
  - Optimized large-scale enterprise databases using MYCAT middleware, achieving higher throughput and reduced query latency.
  - Enhanced data processing efficiency and reliability, supporting high-concurrency payment operations.

Education

University of Auckland, Auckland, New Zealand
Visiting PhD Student

Artificial Intelligence in Industry, Faculty of Engineering

Tongji University, China
PhD

Control Science and Engineering, School of Electronic and Information Engineering

Doctoral Dissertation: Research on Automated Assembly Sequence Planning and Quality Inspection Methods Based on Multi-dimensional Semantic Knowledge Modeling

- The First Prize Scholarship
- The Second Prize Scholarship
- Excellent doctoral dissertation

Tongji University, China
M.S. degree

Control Engineering, School of Electronic and Information Engineering

Master's Dissertation: Research and Application of Production Visualization System based on Augmented Reality in Digital Workshop

- The Phoenix Scholarship

Skills

Machine Learning

Computer Vision

Graph Learning

Generative AI

Python Programming

PROJECT EXPERIENCE

Multimodal Language Model Enterprise Automation Tool Construction and Deployment
2024.12 – 2025.04

Supported by Shanghai Zaixing Testing Technology Co., Ltd – Project Lead

- Designed and deployed an enterprise automation tool powered by a multimodal large language model.
- Integrated corporate knowledge and data to build and run end-to-end workflows, streamlining business operations.

Huayi Energy & Chemical Digital Twin AI Platform
2021.12 – 2022.09

Supported by Shanghai Huayi (Group) Company – Major Participant

- Conducted on-site investigations of the Huayi Chemical Plant to assess operational needs.
- Planned the functions and technical modules for the digital twin AI platform, optimizing factory operations.

Subway Track Area Segmentation and Obstacle Identification
2021.03 – 2023.01

Supported by Shanghai Qnode Technology Co., Ltd – Project Lead

- Utilized 3D point cloud segmentation methods to detect subway track areas.
- Collected and analyzed subway data to build a dataset, and used the SSD algorithm for obstacle identification.

PROJECT EXPERIENCE

Smart Factory: Technology for Logistics Optimization and Collision Avoidance Based on 3D Real-time Localization

2018.01 – 2022.06

National Major Program for International S&T Cooperation: China & Germany – Major Participant

- Managed the overall project and advanced the project schedule.
- Developed a digital twin system for a hydraulic cylinder factory, managing 3D models of equipment.
- Led the development of workshop precise positioning technology based on SLAM.
- Designed AR applications for logistics distribution to enhance operational efficiency and safety.

Additional information

Publications

- Liu, Y.*, **Xia, L.***, Liu, H., Yan, X., Liang, J., Li, H., & Gai, K. (2026). ALM-MTA: Front-Door Causal Multi-Touch Attribution Method for Creator-Ecosystem Optimization. Proceedings of the Fourteenth International Conference on Learning Representations (ICLR 2026). (Under Review, AI Top Conference, Co-first and Corresponding Author*)
- **Xia, L.**, Lu, J., Fan, Y., & Zhang, H. Integrating Augmented Reality and Small Object Detection for Enhanced Logistics Management in Flexible Workshops. Computers in Industry (Under Review) (Impact Factor:9.1, SCI, JCR Q1, Top Journal)
- Liu, Y., Miao, Y., **Xia, L***. Directed Routing Gradient (DRGrad): A Personalized Information Surgery for Multi-Task Learning (MTL) Recommendations. Advances in Neural Information Processing Systems, AAAI 2025 (AI Top Conference, Corresponding Author)
- **Xia, L.**, Lu, J., Lu, Y., Gao, W., Fan, Y., Xu, Y., & Zhang, H. (2024). Semantic knowledge-driven A-GASeq: A dynamic graph learning approach for assembly sequence optimization. Computers in Industry, 154, 104040. https://doi.org/10.1016/j.compind.2023.104040, (Impact Factor:9.1, SCI, JCR Q1, Top Journal)
- **Xia, L.**, Lu, J., Lu, Y., Zhang, H., Fan, Y., & Zhang, H. (2024). Augmented reality and indoor positioning based mobile production monitoring system to support workers with human-in-the-loop. Robotics and Computer-Integrated Manufacturing, 86, 102664. https://doi.org/10.1016/j.rcim.2023.102664, (Impact Factor:11.2, SCI, JCR Q1, Top Journal)
- **Xia, L.**, Lu, J., Zhang, H., Xu, M., & Li, Z. (2022). Construction and application of smart factory digital twin system based on DTME. The International Journal of Advanced Manufacturing Technology, 120(5-6), 4159-4178. https://doi.org/10.1007/s00170-022-08971-1, (Impact Factor:3.2, SCI, JCR Q2)
- Lu, J., **Xia, L***., Zhang, H., & Xu, M. (2022). Research and application of manufacturing enterprise digital twin ecosystem. Computer Integrated Manufacturing Systems, CIMS, 28(8), 2273–2290. https://doi.org/10.13196/j.cims.2022.08.001. (EI Journal, Corresponding Author)
- **Xia, L.**, Lu, J., & Zhang, H. (2020, December). Research on construction method of digital twin workshop based on digital twin engine. In 2020 IEEE International Conference on Advances in Electrical Engineering and Computer Applications (AEECA) (pp. 417-421). IEEE. https://doi.org/10.1109/AEECA49918.2020.9213649 (Conference and Presentation)
- Software Copyright of Tongji Production Monitoring System Based on Augmented Reality (V1.0). Certificate No.0949442, July 2021. (Issued Software)
- Software Copyright of Tongji Part Assembly Induction Prototype System Based on Augmented Reality (V1.0). Certificate No.4749595, September 2019. (Issued Software)
- Software Copyright of Tongji i5OS Experimental Unit Digital Twin System (V1.0). Certificate No.3003770, June 2018. (Issued Software)